

Design of a fully articulated, superbiorimimetic prosthetic arm

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Abstract

This paper presents the comprehensive design of a neurophysiologically informed prosthetic arm, integrating mechanical, electrical, and software subsystems to deliver function and dexterity rivaling the human upper limb. The work begins with a review of current prosthetic technologies, including both commercial devices and recent research innovations, highlighting limitations in motion fidelity, responsiveness, and user adaptability. Most existing systems rely on uninformed surface electromyography (sEMG) signals to extract energy or frequency-based features, which are then used to classify and trigger discrete hand grasps. In contrast, the approach proposed here leverages neurophysiological principles to extract spatial, temporal, and spatiotemporal features that better reflect the motor commands originating from the central nervous system. These features are used to drive a novel actuation scheme based on dielectric elastomer actuators, coordinated to emulate muscle synergies rather than simple independent joint control. The result is a system capable of producing complex, high-degree-of-freedom motions with fewer control signals. The mechanical architecture is designed to match or exceed the force, speed, and dexterity metrics of the biological hand, while remaining lightweight and compliant. This integration of neuroinspired control with biomimetic actuation presents a new paradigm for upper-limb prosthetics, pushing beyond classification-based models toward more naturalistic and adaptive control.

Keywords

Upper limb, Prosthetics, Surface Electromyography, Biomimetic

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1 - Overview

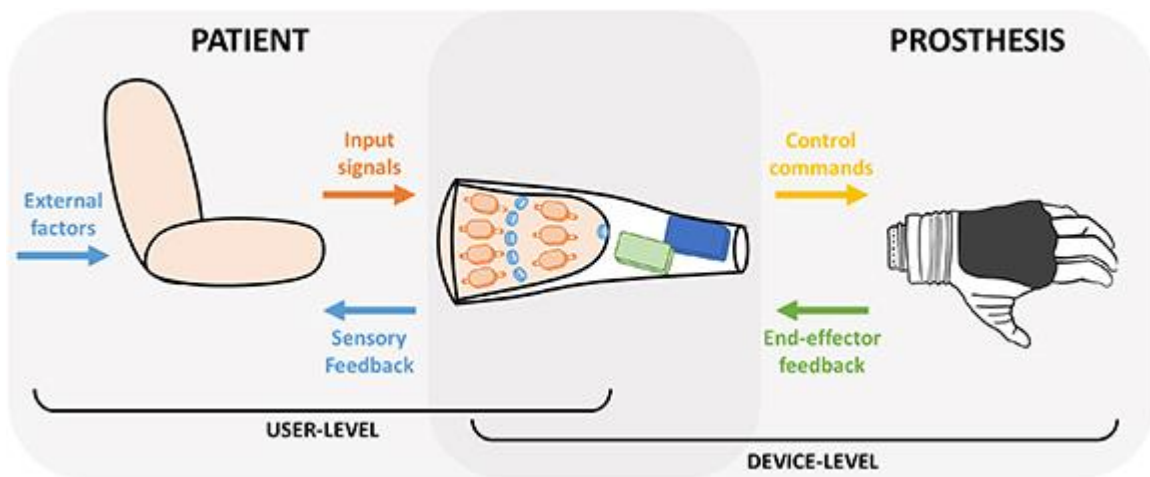
A prosthetic arm is an artificial device designed to replace the function or appearance of a missing upper limb. These devices generally fall into four categories: aesthetic (designed for appearance only), passive (non-powered and capable of limited function), body-powered (controlled by cables or harnesses linked to body movement), and bionic (electromechanical systems that use sensors and actuators to restore motion). This paper focuses exclusively on bionic prosthetics, particularly those controlled by surface electromyography (sEMG) – electrical signals generated by residual muscles in the user's limb. Within this domain, the design targets transhumeral (above the elbow, below the shoulder) prostheses, covering motion of the elbow, wrist, and all five fingers. Prosthetic designs involving the shoulder are not addressed, as they require deeper integration with torso-based musculature and are less common.

The paper is structured to guide the reader from context to implementation. Section 1 reviews existing bionic prosthetic systems, including both commercially available models and designs developed in research settings. Section 2 provides essential biological background on anatomy, biomechanics, and neuromuscular control. Section 3 introduces the proposed system, detailing its mechanical, electrical, and software components. Section 4 compares the performance of this system with current alternatives, and the final section discusses future directions for development. The remaining sections conclude the paper and provide additional technical details.

1.1 Overview of Current Prosthetic Designs

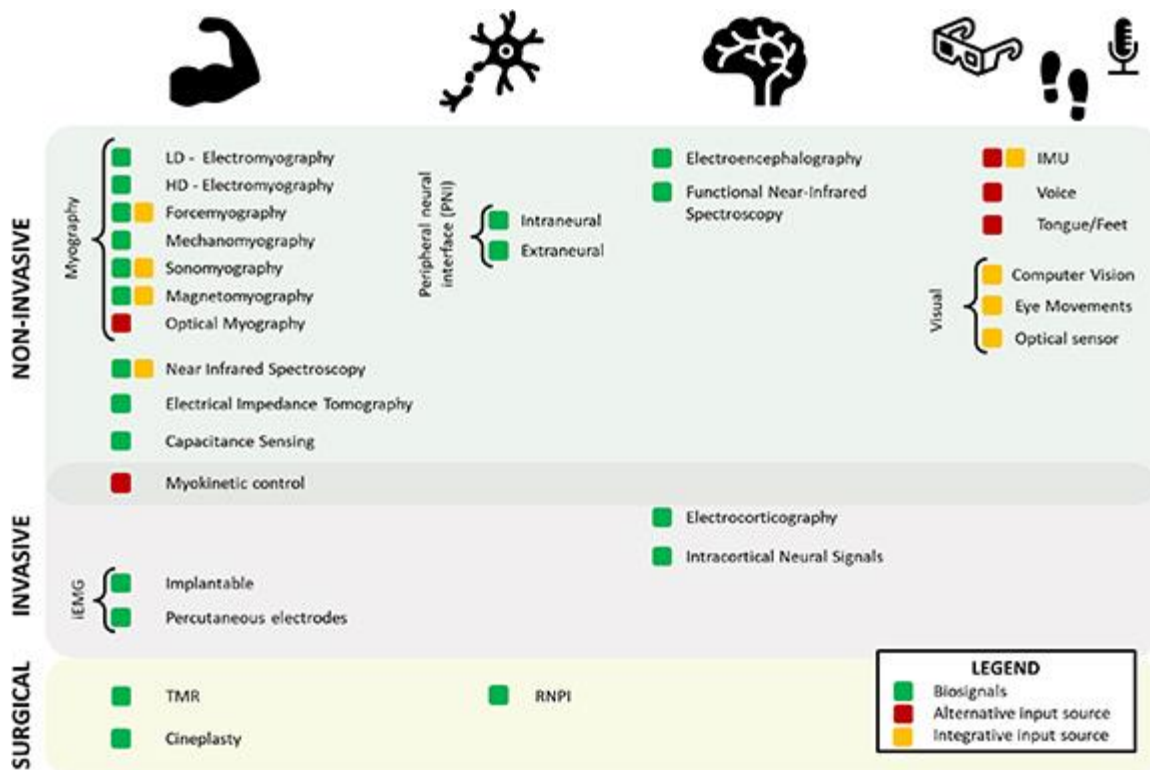
Nearly all modern bionic prosthetic arms (henceforth “prosthetics”) share a common high-level architecture. Each includes a method of attachment to the user, a system for sensing control signals, and an end effector that functions as a robotic hand. *Figure 1* provides a visual overview of this structure. Beyond this basic layout, prosthetic systems vary widely in their attempt to replicate human strength, dexterity, and appearance.

Attachment methods typically fall into one of three categories: (1) over-the-shoulder harnesses, (2) socket-based systems, and (3) osseointegration, where the prosthesis connects directly to the user's bone. The first method is rarely used for transradial amputees except in cases where the amputation is so proximal that socket-based solutions are not viable. Socket systems are the most common due to their ease of removal and relatively low invasiveness, though they require a custom-fitting process performed by trained clinicians. Osseointegration offers improved load-bearing capability and comfort but involves a more invasive surgical procedure.



<https://doi.org/10.1088/2516-1091/acac57>

Figure 2 shows the most common types of control signal inputs, categorized as either non-invasive or invasive. Surface electromyography (sEMG), and its high-resolution variant HD-sEMG, remain the most widely used non-invasive sensing techniques. Invasive methods—including implanted electrodes and nerve interfaces—have shown strong potential but carry the significant drawback of requiring surgery. Additionally, several techniques classified under “surgical” in the figure, such as Targeted Muscle Reinnervation (TMR), do not generate signals themselves but instead amplify other signals by redirecting nerves to more accessible muscle tissue.



<https://doi.org/10.1088/2516-1091/acac57>

doi.org/10.3390/prosthesis6020022 summarizes notable commercial and research-based prosthetic designs developed between 2018 and the present. Among commercial manufacturers, Ottobock and Össur currently hold the largest market shares in the high-end segment, while newer companies aim to offer more affordable alternatives with competitive performance. For example, Psyonic offers one of the lowest-cost commercial bionic hands at around \$8,000 MSRP—still a prohibitive price for many users in low-income regions of the world.

Despite continued innovation, doi.org/10.3390/prosthesis6020022 also shows that current prosthetic designs remain limited. Most systems are underactuated and do not replicate the full number of degrees of freedom found in the human hand. Features such as finger abduction/adduction are often omitted, and complex motions like thumb opposition and flexion are typically merged into a single hybrid movement, reducing the fidelity of control and expressiveness.

1.2 – Common Control Architectures

At an abstract level, prosthetic control systems can be broken down into three main components: (1) Inputs, (2) Controller, and (3) Outputs. Of these, the output stage is generally the most straightforward to implement. It typically consists of an inverse kinematics algorithm or predefined joint trajectories that convert control signals into joint movements based on the prosthetic's mechanical structure.

The input stage is more complex and can include a variety of biosignals such as sEMG, mechanomyography, and neural interfaces. However, for the purposes of this paper, the focus is limited to sEMG as the primary signal. For readers interested in other signal types and their roles in prosthetic control, doi.org/10.1088/2516-1091/acac57 provides a comprehensive overview of alternative input modalities used in both commercial and experimental systems.

sEMG originates from the activity of motor neurons, which send electrical signals to muscle fibers within their motor unit. As individual muscle cells contract, they generate small electrical impulses. These impulses propagate through surrounding tissue, activating nearby cells and creating a cascading effect within the motor unit. Importantly, this does not necessarily mean the entire muscle contracts—only those fibers innervated by the active motor neuron. The summed electrical activity of these fiber groups produces the sEMG signal measurable at the skin surface.

However, sEMG is an indirect representation of muscle force and can be affected by several factors. Signal quality is influenced by skin-electrode interface conditions, body composition, and electrode placement, all of which can distort the recorded signal. Additionally, due to the propagating nature of muscle activation, crosstalk from adjacent muscles is a common and significant challenge in signal interpretation. For a deeper discussion on the origin, behavior, and limitations of sEMG signals, doi.org/10.1002/9781119082934.ch15 serves as a useful technical reference.

Once the sEMG signal is obtained, it must be processed to extract useful information. The first layer of processing is known as feature extraction, which involves reducing the raw signal to a set of numerical descriptors. These can include time-domain features such as the integrated EMG (IEMG) or zero-crossing count, as well as frequency-domain features like the Fourier spectrum. These features are used as input to classifiers or control algorithms. [Doi.org/10.1109/86.481972](https://doi.org/10.1109/86.481972) provides a comprehensive list and explanation of commonly used features in sEMG-based systems.

In more advanced systems, sensor fusion techniques are employed to improve control robustness and reliability. These approaches combine sEMG with additional sensing modalities such as inertial measurement units (IMUs) or force sensors to provide a more complete representation of user intent. By integrating multiple data sources, these

systems can better handle signal variability and environmental disturbances.

[Doi.org/10.1016/j.robot.2014.08.012](https://doi.org/10.1016/j.robot.2014.08.012) offers a detailed overview of sensor fusion strategies as applied to prosthetic control. In a similar vein, PaWFE is an excellent example of feature extraction doi.org/10.3389/fnbot.2019.00074.

Given the complexity of sEMG signals and the limited understanding of their physiological origins, many researchers rely on machine learning as the core of the control system. From a high-level perspective, this is a reasonable choice—machine learning algorithms excel at identifying complex patterns in data. These approaches typically operate by learning statistical relationships between extracted features and desired movements, often using classifiers, regressors, or neural networks.

This paper does not focus on developing or evaluating machine learning algorithms, and instead refers readers to the extensive literature on the subject. Numerous high-quality resources exist that cover both the theoretical foundations (see below) of machine learning and its practical applications (See below) in prosthetic control.

Theoretical Info:

doi.org/10.1007/s11263-006-9794-4

doi.org/10.1093/pan/mpt019

doi.org/10.48550/arXiv.1802.01396

doi.org/10.1109/TNNLS.2015.2476659

doi.org/10.1109/TNSRE.2014.2303394

<http://cbcl.mit.edu/publications/ps/rlsc.pdf>

Practical info:

doi.org/10.1109/TNSRE.2017.2699598

doi.org/10.1109/TNSRE.2008.2006212

doi.org/10.48550/arXiv.2312.09418

Ninapro – doi.org/10.1186/s12984-019-0488-x

Sensor fusion + Ninapro = >90% accuracy – doi.org/10.3389/fbioe.2021.548357

One noteworthy contribution to this area is the NinaPro database, a widely used benchmark in the prosthetics research community. Most machine learning-based controllers focus on identifying a limited set of functional hand grasps—such as a fist, pinch, or pencil hold—by recognizing an sEMG pattern. Importantly, these approaches are generally not concerned with identifying individual muscle activations; instead, they classify whole-hand gestures as discrete units. In commercial prosthetic systems, the number of available grasp types is typically small, often ranging from 10 to 20 distinct motions, with the most advanced systems supporting up to around 30.

NinaPro extends this concept by defining and supporting 41 unique grasp types, each backed by physiological and functional justifications. While the supporting publications offer excellent insight, the broader trend toward increasing the number of distinct grasp classes points to a natural end goal: a fluid continuum of hand motions that can represent any hand posture, not just a finite set of pre-defined classes. However, this vision is fundamentally at odds with most conventional machine learning methods, which are built around discrete classification. Few current approaches incorporate fuzzy logic or continuous-state models that would allow for nuanced transitions between similar hand shapes—a limitation that constrains how naturally a user can interact with their environment.

Publications with the Ninapro dataset typically have very high accuracies. The next section is an attempt to replicate and improve the accuracy.

1.3 – Application of Machine Learning

The data collected for this section is from a script written in MATLAB. Due to hardware constraints, the 2kHz data has been downsampled to 100Hz.

Following the suggestions from recent publications, a machine learning approach using a Kernel-Based Regularized Least Squares was considered with various kernels. Four candidate kernels were selected: $\exp(-\chi^2)$, Radial Basis Function (RBF), Laplacian and Linear. The $\exp(-\chi^2)$ kernel may be more appropriate for histogram-like non-negative feature representations, such as the one created from sEMG data.

$$K_{\exp-\chi^2}(\mathbf{x}, \mathbf{y}) = \exp\left(-\gamma \sum_{i=1}^n \frac{(x_i - y_i)^2}{x_i + y_i}\right) \text{ for } \gamma > 0$$

$$K_{\text{RBF}}(\mathbf{x}, \mathbf{y}) = \exp\left(-\gamma \|\mathbf{x} - \mathbf{y}\|^2\right) \text{ for } \gamma > 0$$

$$K_{\text{Laplace}}(\mathbf{x}, \mathbf{y}) = \exp(-\gamma \|\mathbf{x} - \mathbf{y}\|) \text{ for } \gamma > 0$$

$$K_{\text{Linear}}(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$$

Using the data from Subject 1 of the first NinaPro database and the mDWT features extracted from that data, the performance of the kernels is listed in Figure 1. One especially important item to notice is that the Linear kernel performed significantly worse than others. This hints at the non-linear nature of the data.

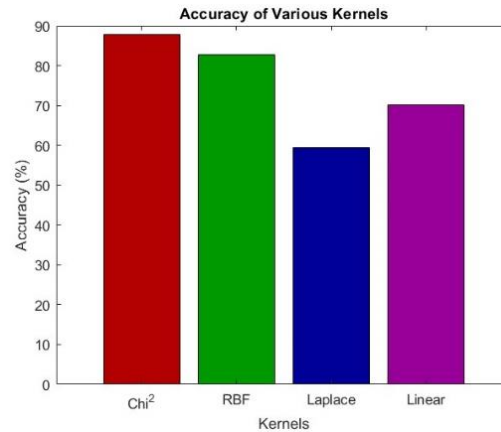


Figure 1: Kernel Performance

More features usually improves the classification accuracy, but prosthetics require real-time classification, so the number of features cannot be expanded endlessly. As such, even though there are numerous candidate feature extraction methods and an even larger number of combinations, this report is limited to just the three popular feature extraction methods discussed previously.

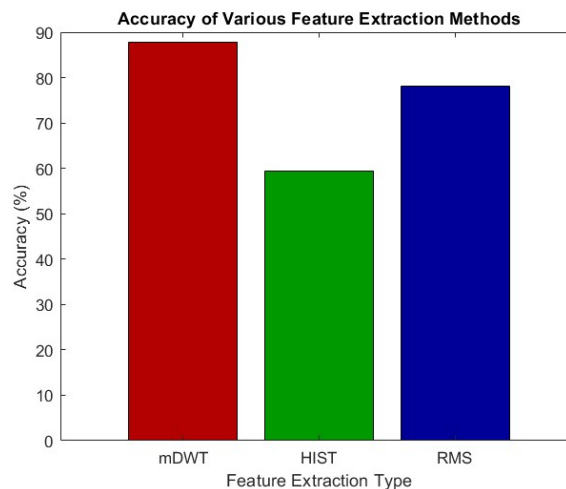


Figure 2: Feature Performance

Due to the large number of zeros created by the HIST feature extraction method, unreasonably low performance was displayed. It needs to be revisited to determine how to better approach the data.

Currently, MATLAB offers an app called the Classification Learner that will train numerous models based on common machine learning algorithms. Figure 3 shows a table of all models that achieved above 80% accuracy. Of them, none were more accurate than the 89.2% achieved with the $\exp(-\chi^2)$ / mDWT model, which suggest that the unusual kernel is more suitable for this specific task than the general models that MATLAB presents.

An interesting occurrence that may be noteworthy is that the general models trained to a much higher accuracy using the testing data from the first cross-validation set, but then performed much worse than the presented values upon validation. Since this is also the validation set that the $\exp(-\chi^2)$ performed the worse at, this suggest that the first two repetitions for each task in the dataset is numerically different than the remainder.

MATLAB Classification Learner - Highest Performance			
Learner	Accuracy	Learner	Accuracy
Cubic SVM	88.1	Bagged Trees	85.5
Quadratic SVM	87.6	Wide NN	85.5
Weighted KNN	87.4	Medium NN	85
Medium KNN	86.8	SVM Kernel	84.4
Cubic KNN	86.8	Linear Discriminant	83.3
Medium Gaussian SVM	86.6	Narrow NN	83.3
Subspace KNN	86.6	Subspace Discriminant	83.1
Linear SVM	86.2	Trilayered NN	82
Cosine KNN	86.1	Bilayered NN	80.6

Figure X: MATLAB Classification Learner Models

MATLAB also offers two Neural Net apps suggested for classification, but they both performed very poorly on this dataset.

From this analysis, several conclusions can be drawn. First, sEMG data is non-linear. This restricts any attempts at processing to non-linear processes. Next, sEMG data contains some underlying quality that results in generalized machine learning approaches under-performing. The $\exp(-\chi^2)$ kernel is most well-known for comparing histograms which arise from an object having a distinct pattern to the signal/data representation. Considering the underlying anatomy and periodicity of the nervous system may be why the kernel works well. The window (100ms) is wide enough to capture several activations of a firing nerve and subsequent spread as nearby nerves fire in response.

Yet, it's the mDWT rather than the HIST feature extraction that performs best. This initial response is because of poor notation. The HIST divides up the window while mDWT extracts information relating to the signal's form and periodicity, much like a Fourier transform. In this sense, the mDWT extracts the information that is useful for the kernel better than that of HIST.

In the end though, it's exactly this property of the signal that causes problems with cross-compatibility between individuals. Taking the estimated parameters from Subject 1 and trying to classify the movements of Subjects 2 through 5 results in a significant decrease in performance. On one hand, the results are much better than random guessing (5.56%), but it's far too low to be usable in the daily life of a prosthetics user where incorrect classifications are heavily disfavored.

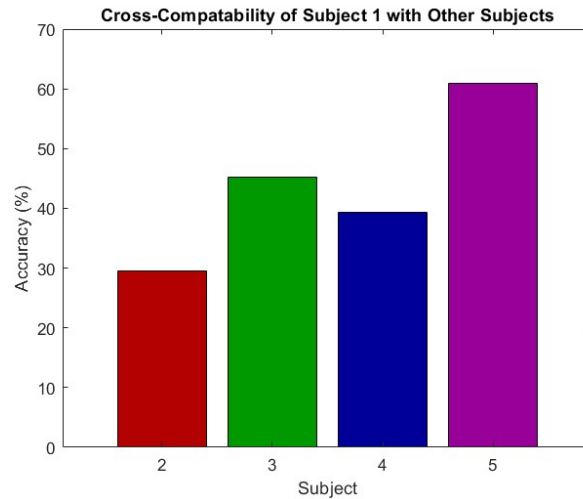


Figure 4: Cross compatibility

1.4 – Motivation for a Neurophysiological Approach

Since the human hand operates with near-perfect reliability, the control signal is fundamentally discernible. Yet, despite extensive exploration of black-box methods from many researchers, classification often falls short of 90% accuracy for training data based on a limited number of grasps. Their performance tends to degrade further when tested on validation or new data. One contributing factor is the presence of numerous non-linearities in the neuromuscular and sensor pathways, which reduce the effectiveness of linear kernels and challenge the generalizability of even non-linear controllers. These complexities must be explicitly accounted for rather than treated as noise.

One promising direction is to introduce biologically informed constraints into the control system. For example, doi.org/10.48550/arXiv.2312.09418 shows that incorporating physical constraints can reduce error and improve system accuracy. Rather than treating the human body as an arbitrary input-output system, it may be more effective to incorporate domain knowledge of musculoskeletal dynamics and neural control to guide controller design.

Additionally, there is a high degree of inter-user variability in sEMG signals, driven by differences in anatomy, muscle density, skin impedance, and motor learning strategies. While some studies (e.g., <https://doi.org/10.3390/app12062962>) report overly narrow or generalized conclusions, the evidence increasingly supports the idea that a universal controller may be unrealistic. As will be discussed in the following sections, sEMG signals are shaped by anatomy, electrode placement, and neuromuscular patterns—all of which vary significantly across individuals. An approach that explicitly acknowledges and adapts to this variability may offer a more robust and personalized path forward.

2 Necessary Biological Context

While machine learning and other black-box methods are popular in prosthetic control, they are progressing slowly because they ignore the underlying structure and constraints of the human motor system. To build more robust and effective prosthetics, future approaches must go beyond just mapping inputs to outputs—they need to adopt a "clear-box" strategy where the inner workings of both the biological and control systems are well understood. This section provides an overview of the anatomy, forces, physiology, and neural control involved in upper-limb movement, laying the groundwork for more biologically-informed prosthetic designs. Additional resources are included if more knowledge than this overview is necessary.

2.1 *Anatomy of the Human Arm and Hand*

To better understand the control signals generated by the nervous system, it is helpful to first understand how a healthy arm functions. Since anatomy is a well-established field, this paper will refer readers to doi.org/10.1055/b-004-135650, a textbook chapter that provides a detailed overview. In addition, because visual guides can make learning anatomy easier, a series of YouTube videos have been selected and are listed below. It is recommended to watch them in order for a structured introduction.

- Hand Muscles - https://www.youtube.com/watch?v=B6KCGR7j_Tw
- Forearm flexor muscles – <https://www.youtube.com/watch?v=jtKZA6sHdAY>
- Forearm extensor muscles - <https://www.youtube.com/watch?v=WkQjUNdP2JM>
- Hand Anatomy - <https://www.youtube.com/watch?v=zyl6eoU-3Rg>
- Finger Flexor Pulley Anatomy - https://www.youtube.com/watch?v=mNnaZRxC_Kk
- Forearm, Wrist, Hand Anatomy - <https://www.youtube.com/watch?v=Cz3-mIsJtdE>
- Elbow Anatomy - <https://www.youtube.com/watch?v=3km9pkS75pQ>
- Shoulder Anatomy - <https://www.youtube.com/watch?v=daWOGS7fbzA>

In the context of prosthetic control, it is important to note that surgical interventions are sometimes used to improve signal quality. Because amputation already involves surgery, it is possible to perform additional procedures that enhance the available control signals. One example is Targeted Muscle Reinnervation (TMR), described in doi.org/10.1055/s-0035-1544166, which reroutes nerves from the amputated limb to remaining muscles,

effectively creating new signal sources with improved signal-to-noise ratios (SNR) and additional control dimensions. However, requiring auxiliary surgeries like TMR introduces additional risks and limits the number of users who may be eligible. Whenever possible, non-surgical methods are preferred to maximize accessibility.

2.2 – Biomechanical Forces, Moments, and Interfaces

The structure of the upper limb directly determines the biomechanical forces and moments it can produce. For a prosthetic system to be truly biomimetic, it should aim to match or exceed the strength and movement capabilities of a healthy human arm. Because of this, understanding the limb's dynamic behavior is a natural next step after reviewing its anatomy.

To guide this discussion, this paper will reference the joint system described in doi.org/10.1016/j.jbiomech.2004.05.042, which outlines the major degrees of freedom and mechanical relationships between joints. It is assumed that position measurements are obtained using inertial measurement units (IMUs) placed according to the methods described in doi.org/10.1016/j.jbiomech.2024.112225. This provides a practical framework for modeling motion and for designing prosthetic systems that accurately replicate natural joint dynamics.

Although it is mechanically a simple hinge joint, the elbow must withstand and produce significant forces during normal use. The magnitudes and types of forces involved are discussed in detail in the papers below, which cover both static and dynamic loading conditions.

- Elbox overview - doi.org/10.1007/978-3-030-14455-5_4
- Elbow Force – doi.org/10.1371/journal.pone.0218553
- Elbow Moment - [doi.org/10.1016/0021-9290\(81\)90048-8](https://doi.org/10.1016/0021-9290(81)90048-8)
- Elbow Moment - doi.org/10.1115/1.3138507
- Angle of max elbow force - doi.org/10.1007/BF00235099

The wrist presents a much more complex structure. It consists of seven small carpal bones arranged in two rows, forming multiple interconnected joints. This complexity results in a highly flexible but mechanically intricate interface, with motions that are not easily approximated by simple rotational models. Detailed information on wrist biomechanics, including joint interfaces, relative bone motions, and force transmission pathways, can be found in doi.org/10.1002/jor.24435, doi.org/10.103390/life12020188, [doi.org/10.1016/s0021-9290\(97\)00026-2](https://doi.org/10.1016/s0021-9290(97)00026-2), [doi.org/10.1016/0363-5023\(88\)90189-x](https://doi.org/10.1016/0363-5023(88)90189-x).

Additionally, [doi.org/10.1016/0003-6870\(94\)90057-4](https://doi.org/10.1016/0003-6870(94)90057-4) provides quantitative data on the total force-generating capability of the wrist when considered as a unit.

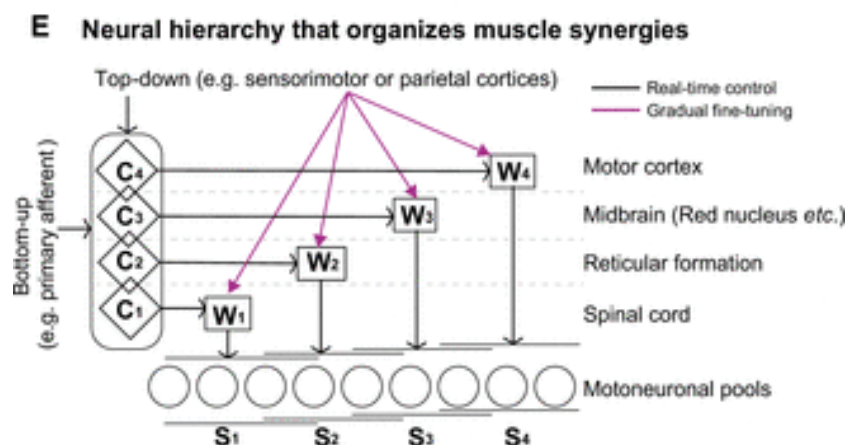
Strength data for the fingers is most commonly reported in terms of overall grip strength, such as the values summarized in doi.org/10.1002/jcsm.12610. However, it is important to note that not all fingers contribute equally to gripping force. Differences in individual finger strength are detailed in [10.1080/00140139.2011.620176](https://doi.org/10.1080/00140139.2011.620176), which provides breakdowns by digit and grip type.

[Insert summary table of moments and moment arms by joint]

2.3 – Muscular and Physiological Function

As discussed in Section 2.1, muscles are organized into muscle synergies—groups of muscles that are activated together to produce functional movements. However, the motor unit remains the smallest unit of neural control. As force demands increase for a specific movement, nearby motor units are recruited in a somewhat predictable sequence, a phenomenon known as motor unit recruitment. This recruitment pattern, along with other evidence, suggests that certain motor units are correlated based on the motor commands they receive.

Using the framework described in doi.org/10.1152/jn.00625.2019, the recorded sEMG signal can be decomposed into two important matrices. The first, W , represents the spatial coordination of muscle synergies across multiple muscles. The second, C , represents the temporal coupling of activations across these synergies. Figure 4 provides a visualization of this decomposition. See doi.org/10.1016/j.neuroscience.2023.01.017 and doi.org/10.1088/1741-2552/ac5150 for more info.



doi.org/10.1152/jn.00625.2019

However, the synergy matrix W is not static; it is time-varying, and is more accurately expressed as $W(x, t)$. Time variations arise due to factors such as muscle fatigue, strengthening through repeated use, or long-term learning processes often referred to as "muscle memory." To simplify the problem, this paper proposes decomposing $W(x, t)$ into two components: a static matrix $W_a(x)$ that captures the baseline organization of muscle synergies, and a time-varying matrix $W_b(x, t)$ that accounts for dynamic changes over time. In the context of prosthetic design, both W_a and W_b would require calibration. Initially, W_a could be calibrated during setup and then slowly updated by monitoring signals during periods of low activity, such as sleep, or during movements performed at moderate force levels, where muscles behave more linearly. W_b could be updated by tracking long-term trends during repetitive actions, capturing gradual shifts related to fatigue or retraining.

$$W = W_a(x) + W_b(x, t)$$

Similarly, the command structure represented by matrix C is also nonlinear and time-varying. As will be discussed further in the next section, $C(t)$ can likewise be decomposed into two components: $C_a(t)$, representing the user's intended actions and forces, and $C_b(t)$, capturing temporal variations such as those caused by learning, physiological delays, or psychological refractory periods.

$$C = C_a(t) + C_b(t)$$

Overall, this framework separates the control problem into two layers: a predictable baseline that defines how an action should normally be performed, and a dynamic correction that accounts for temporal effects and adaptive changes over time.

2.4 – Neural Command Strategies: Intermittent and Hierarchical Control

Sections 2.1 and 2.2 discussed the physical structure of muscles and how they generate motion in the upper limb. To extend this understanding to control design, it is useful to first examine how muscles are typically modeled. A widely used representation is Hill's muscle model, which describes muscle behavior using a linear ordinary differential equation (ODE). If muscle function were purely mechanical and linear, a corresponding control system could be designed with simple linear assumptions.

However, real muscle function is fundamentally nonlinear, driven by chemical signaling, motor unit recruitment, and complex intracellular dynamics that Hill's model does not capture. The nervous system has evolved to control these nonlinearities, meaning that neural control signals are not easily modeled as smooth, continuous functions. Even if a prosthetic joint exhibits simpler mechanical properties, the control signals intended to drive it still reflect the complexities of a healthy limb.

One important feature of neural motor control is its intermittent and sparse nature. Rather than continuously micromanaging muscle fibers, the brain sends brief, high-level commands that are interpreted and expanded by lower-level neural circuits. This sparse control architecture reduces cognitive load but introduces challenges for prosthetic systems, which must be able to interpret both discrete control commands and continuous sensory feedback. Controllers must therefore be designed to extract meaningful patterns from both time-domain and frequency-domain characteristics of the input signals.

The complexity of the neural control structure also means that the location of signal measurement significantly affects the properties of the recorded signals. Signals obtained from EEG (brain-level) differ fundamentally from those captured at the muscle (EMG) or spinal (invasive electrode) levels. Even among EMG signals, surface recordings (sEMG) differ from intramuscular measurements, due to variations in signal filtering, crosstalk, and proximity to active motor units.

Another essential characteristic of the nervous system is its high adaptability. Neural control systems continuously adjust through learning, sensory feedback, and even long-term physiological changes. As a result, any prosthetic control system must be able to accommodate time-varying control signals rather than assuming a static user model.

See the following documents for more details and elaboration of the intermittent and hierarchical control model

doi.org/10.3389/fncom.2013.00012

doi.org/10.1007/s00422-010-0416-4

doi.org/10.1007/s00422-014-0587-5

doi.org/10.3389/fncom.2013.00055

https://en.wikipedia.org/wiki/Psychological_refractory_period

https://en.wikipedia.org/wiki/Tetanic_contraction

See the following documents for more detail about learning and adaptation.

doi.org/10.1007/978-3-642-28572-1_23

doi.org/10.3389/fncom.2013.00055

3 System Design

This section outlines the proposed design for a prosthetic system intended to closely mimic the structure and dynamics of the human upper limb. The overall project is divided into major subsystems, organized according to the work breakdown structure (WBS) shown in Figure X. Each of the following subsections provides a general overview of the design approach for its respective subsystem, with references included where appropriate to clarify key decisions and support the design direction.

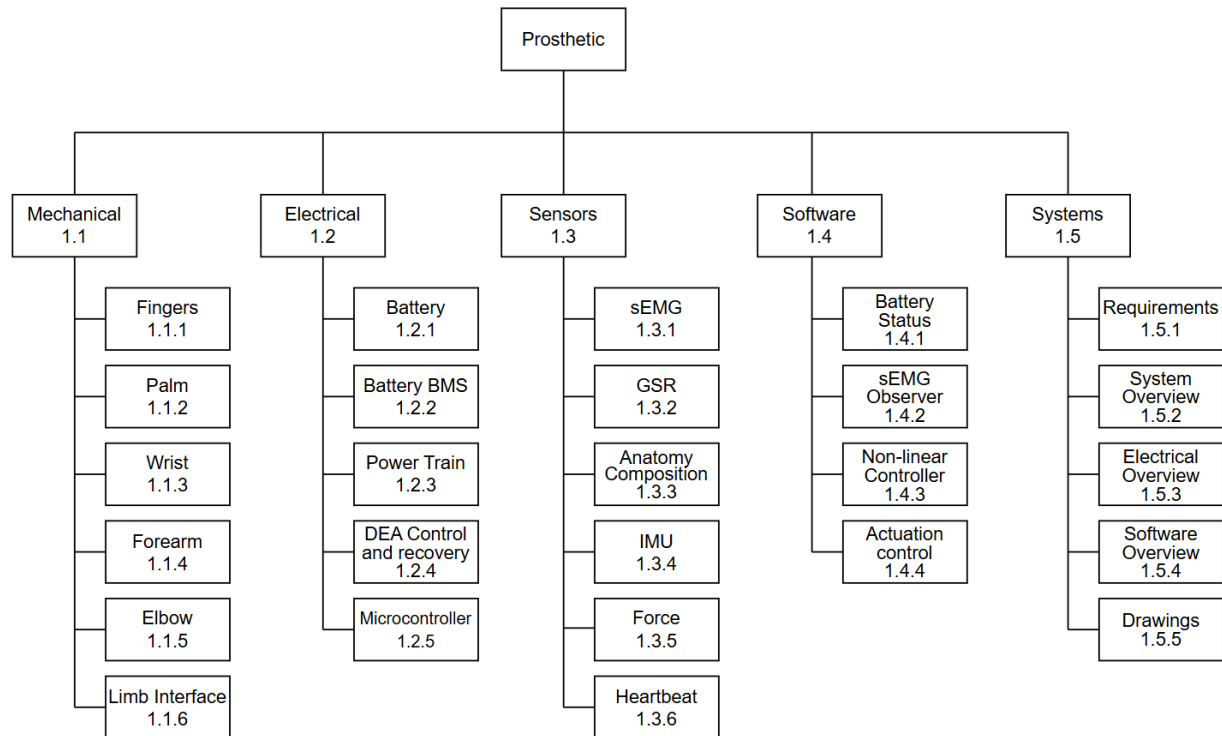


Figure X: Work Breakdown Structure

3.1 – Overall System Architecture

In order to better define the project scope, essential parameters have been established and divided into two categories: Requirements and Engineering Goals. Requirements are statements that include the word "shall" and define the minimum performance or functionality necessary to achieve a viable product. These represent non-negotiable criteria that must be met. The second category, Engineering Goals, includes statements with the word "should." These goals provide direction for the design and list features that, while not strictly required, would enhance the system's competitiveness and overall performance. A full list of Requirements and Engineering Goals is provided in Table

X and X. User verification testing will be performed using the procedures described in doi.org/10.3389/frobt.2023.1292632 to confirm that the Requirements have been met.

Requirements				
ID	Name	Description	Threshold Value	Relevant Research
1	SYS-1	The system shall comply with sections of IEC 60601 that are relevant to a prosthetic arm.	---	---
2	SYS-2	The system shall comply with IEC 60529 to the IP67 rating.	IP 67	---
3	SYS-3	The system shall not cause electrical, thermal or mechanical harm to the user.	---	---
4	SYS-4	The system shall operate in -40degF to 105degF temperatures without failure.	---	---
5	MECH-1	The hand shall have five independent digits.	---	---
6	MECH-2	All digits shall be able to extend/flex and abduct/adduct at least as much as a normal human hand.	See Table X	---
7	MECH-3	The thumb-equivalent digit shall be able to oppose/repose at least as much as a normal human hand.	See Table X	---
8	MECH-4	The wrist shall be able to supinate/pronate, flex/extend and abduct/adduct at least as much as a normal human wrist.	See Table X	---
9	MECH-5	The elbow shall be able to flex/extend at least as much as a normal human elbow.	See Table X	---
10	MECH-6	The system shall generate force at least as much as a normal human limb.	See Table X	---
11	MECH-7	The system shall be able to incorporate various lengths of transradial surgery.	See Table X	---
12	MECH-8	The system shall weigh as much as or less than a normal human hand.	See Table X	---
13	ELEC-1	The system batteries shall not show a degregation of more than 20% over 730 charge-discharge cycles at 90% depth of discharge, or two years of use, whichever comes first.	---	---
14	ELEC-2	The system shall be able to provide power for 14 hours under conditions equivalent to moderate daily activity.	---	---
15	ELEC-3	The system shall allow for a swappable auxiliary battery with a capacity of at least 2 additional hours under moderate daily activity.	---	---
16	SW-1	The system shall respond faster than 250ms.	---	---
17	SW-2	The system shall have an accuracy greater than 90%.	---	---

Table X - Requirements

Engineering Design Goals				
ID	Name	Description	Threshold Value	Relevant Research
18	EG-SYS-1	The system should have a primarily picture-based user manual.	---	---
19	EG-SYS-2	System calibration should take less than one hour when the user is assisted with trained personnel.	---	---
20	EG-MECH-1	The system should be fully enclosed.	---	---
21	EG-MECH-2	The system should be limited such that the user is not harmed by it's speed nor acceleration.	---	---
22	EG-MECH-3	The system should use Dielectric Actuators (DEAs) as the primary means of flexion.	---	---
23	EG-MECH-4	The system should have an adjustable attachment point.	---	---
24	EG-MECH-5	The user interface should be antibacterial.	---	---
25	EG-MECH-6	The user interface should show no degregation when cleaned with hot water and soap.	---	---
26	EG-ELEC-1	The system should have a variant with batteries that have a capacity that supports at least 16 hours under heavy daily activity.	---	---
27	EG-ELEC-2	The system sensitivity should be high enough to not require additional surgery or special surgical techniques during amputation.	---	---
28	EG-SW-1	The system should extract features with a neurophysiological relevancy.	---	---
29	EG-SW-2	The system should report basic health information.	---	---
30	EG-SW-3	The system should report equipment health.	---	---

Table X – Engineering Design Goals

Below is some of the information needed to finish Table X and X. The rest of the information can be found in the summary table found at the end of Section 2.2.

Table 1. Ranges of motion for the movements of each joint [29].

Joint	Types of Movement	Ranges of Motion (ROM)
Elbow	Flexion	0–150°
	Extension	0–10°
	Pronation	0–90°
	Supination	0–90°
Wrist	Flexion	0–60°
	Extension	0–60°
	Radial deviation	0–30°
	Ulnar deviation	0–40°
CMC of digit I	Abduction	0–70°
	Flexion	0–50°
	Extension	ND
	Opposition	NQ
MCP of digit I	Flexion	0–50°
DIP of digit I	Flexion	0–80°
	Extension	0–20°
MCP of digits II, III, IV, V	Flexion	0–90°
	Extension	0–30°
	Abduction	ND
	Adduction	ND
PIP of digits II, III, IV, V	Flexion	0–100°
DIP of digits II, III, IV, V	Flexion	0–90°

I: thumb, II: index, III: middle, IV: ring, V: pinky. CMC: carpometacarpal, MCP: metacarpophalangeal, DIP: interphalangeal, PIP: proximal interphalangeal, NQ: not quantifiable, ND: not determined.

<https://doi.org/10.3390/prosthesis6020022>

3.2 – Mechanical Subsystem

The fundamental goal of a prosthetic arm is to restore functional movement, so the mechanical design process began with a survey of actuator technologies commonly used in prosthetic research. doi.org/10.1002/adma.201704407 and doi.org/10.1016/j.ijmecsci.2023.108758 provide useful references for this survey, highlighting a range of actuator types and their associated performance characteristics.

A personal interview with the technical director of the TALOS exoskeleton project ([https://en.wikipedia.org/wiki/TALOS_\(uniform\)\)](https://en.wikipedia.org/wiki/TALOS_(uniform))), Joe Register, suggested that most traditional actuators have power requirements that are too high to be viable for long-term viability. High energy consumption was named as the primary limitations in TALOS. One newer actuator type that may address this challenge is the Dielectric Elastomer Actuator (DEA). Due to their ability to recycle energy during motion, DEAs offer significantly improved efficiency compared to conventional motor-driven systems.

In addition to efficiency, DEAs exhibit several properties that make them highly attractive for prosthetic applications. As described in doi.org/10.1016/j.ijmecsci.2023.108758, doi.org/10.1080/00222348.2012.659617 key advantages include low power consumption, high specific strength (strength per unit weight), high volumetric strength (strength per unit volume), and long cycle life. Furthermore, DEAs can be easily combined in series and parallel arrangements to customize force output and mechanical behavior to match task-specific requirements.

Beyond actuator choice, achieving high biomechanical fidelity is critical to user acceptance and system functionality. Biomimicry allows users to reuse motion patterns and control strategies learned during natural hand use. doi.org/10.48550/arXiv.2309.06440 demonstrates that prosthetic designs with low levels of biomimicry, while capable of sophisticated manipulation tasks such as in-hand object rotation, often require additional sensors and complex control algorithms to compensate for mechanical limitations. In contrast, a mechanically biomimetic hand simplifies control demands and appears more natural to the user—an aspect frequently reported as important for psychological and social acceptance.

A high degree of freedom (DOF) is another critical design target. As supported by findings in doi.org/10.1038/s41467-025-56352-5 and doi.org/10.1038/s41467-021-27261-0, users are accustomed to fine, independent motions of a healthy hand. Missing degrees of freedom are often perceived as jarring and frustrating, forcing users to adopt compensatory motions that conflict with their ingrained motor patterns.

Bringing these considerations together, this paper will use the design process described in doi.org/10.1109/ICRA.2016.7487528 as a baseline. The process would be extended to the elbow. However, this paper also proposes specific optimizations to the structural supports for improved manufacturability. While 3D-printed bones offer geometric flexibility, they have inferior material properties and present costly challenges for machining from stronger metals. Therefore, the mechanical structure will be adapted to maintain necessary fidelity while ensuring that components can be manufactured efficiently from robust materials.

3.3 – Electrical Subsystem

The electrical system must be robust, reliable, and capable of supporting many hours of continuous operation. Figure 3 shows an overview of the electrical subsystem connections within the work breakdown structure (WBS), while Figure 4 provides a more detailed breakdown of component relationships.

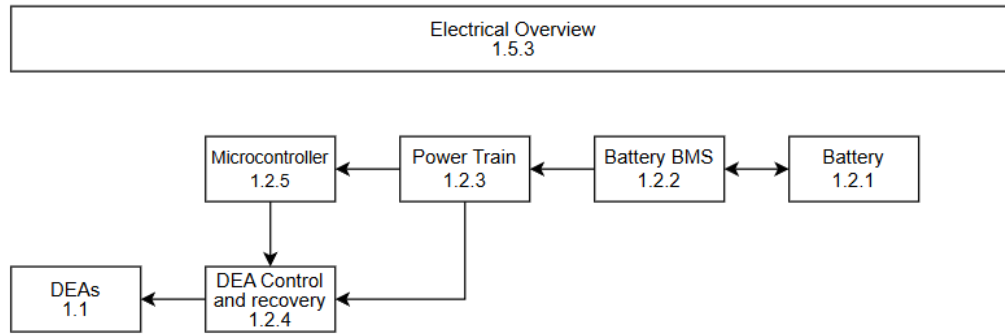


Figure X – Electrical Overview

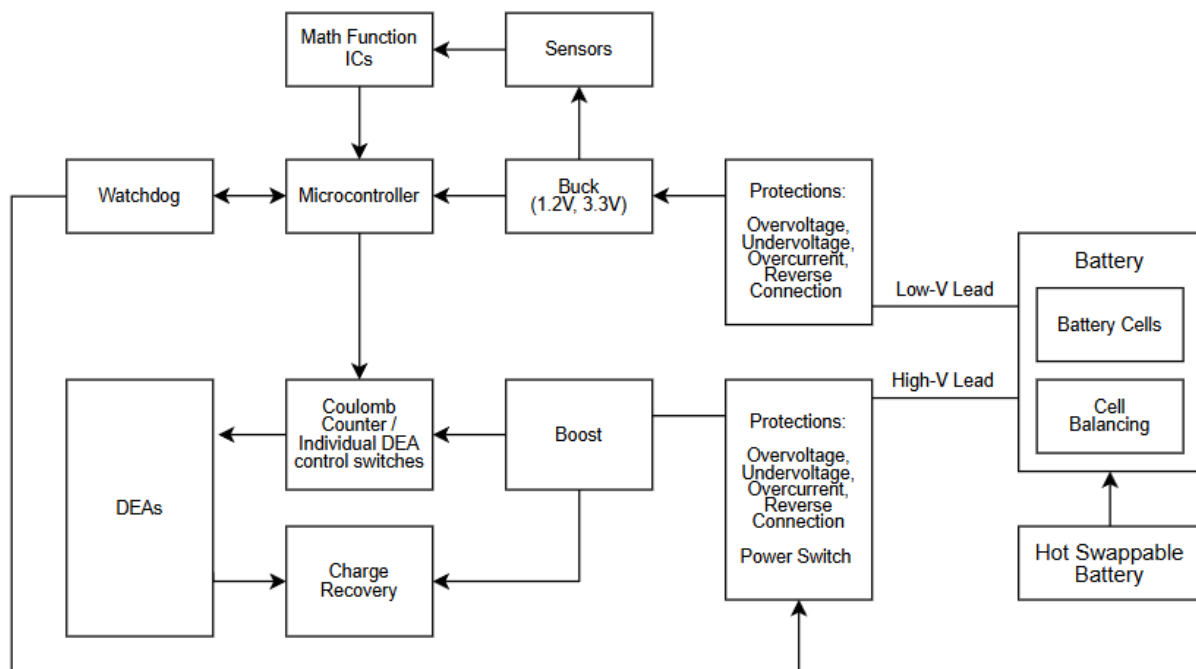


Figure X – Expanded Electrical Overview

Subsystem 1.3.1 – Battery

The battery must be appropriately sized for the prosthetic's use case, balancing energy capacity, weight, and cycle life. Battery chemistry and construction play a major role in meeting these requirements. doi.org/10.3390/batteries11040127 discusses different battery chemistries (see Table 3) and major aging mechanisms. Given a target of approximately one charge cycle per day for three years, there is little need for extremely high-cycle-life batteries. Instead, minimizing weight through high energy density is a priority, especially since dielectric elastomer actuators (DEAs) are relatively low-power devices.

This suggests that newer battery chemistries, which offer substantially higher gravimetric energy densities, may be ideal. Lithium-sulfur (Li-S) batteries, discussed in

doi.org/10.3390/batteries11030089 and <https://www.lis.energy>, appear particularly promising. If Li-S is unavailable or unproven for the application, an advanced version of a more mature chemistry, such as lithium-ion, would be viable.

Table 8. Comparison of lithium-ion battery chemistries.

Application	Chemistry Type	Energy (Wh/kg)	No. of Charge/Discharge Cycles	Heat Resistance	Power Output	Safety Measures	Price Point	Aging Factors	Ref.
EVs, Energy Storage	Lithium Iron Phosphate (LFP)	90–160	2000–5000	Highly Stable at High Temps	Moderate	Safe	Moderate	Volume Expansion of Electrodes, SEI Development	[323,324,325,326]
EVs	Nickel Cobalt Aluminum (NCA)	200–260	500–1000	Needs Better Cooling	High	Improved with Battery Management	Higher	Electrolyte Breakdown, SEI Growth	[327,328,329,330]
Electric Tools, Power Storage	NMC	150–220	1000–2000	Sensitive to High Temps	High	Moderate	Moderate-High	Cathode Degradation, SEI Layer Expansion	[331,332,333]
High Power, Fast Charging EVs	Lithium Titanate (LTO)	80–120	7000–20,000	Stable at All Temps	Very High	Extremely Safe	Higher	SEI Growth, Lithium Plating, Electrolyte Aging	[334,335,336,337]
Portable Electronics	Lithium Cobalt Oxide (LCO)	150–240	500–1000	Overheating Issues	Moderate	Moderate	High	SEI Expansion	[338,339,340]
Next-Gen EVs, Consumer Electronics	Solid-State Lithium-Ion	300+ (Potential)	>10,000	Non-Flammable	High	Extremely Safe	High	Reduced Side Reactions, Solid Electrolyte Interface	[341,342,343,344,345]
Aerospace, EVs	Lithium–Sulfur (Li–S)	400–600	100–500	Stable at Low and Moderate Temps	Moderate	Moderate	Low to Moderate	Lithium Degradation, Electrolyte Decomposition	[346,347,348,349,350,351,352]
EVs, Energy Storage Systems	Lithium–Air (Li–Air)	1000+ (Theoretical)	100–500 cycles	Susceptible to Contamination	High	Low	High	Lithium Degradation	[353,354,355,356,357]

<https://www.mdpi.com/2313-0105/11/4/127>

A survey of existing commercial prosthetic arms shows typical battery capacities of around 2 Ah, yielding operational lifetimes of roughly 8 hours (see Figure X). Since DEAs require less power than conventional motor-driven systems, similar-sized batteries would likely provide even longer operational times without additional weight.

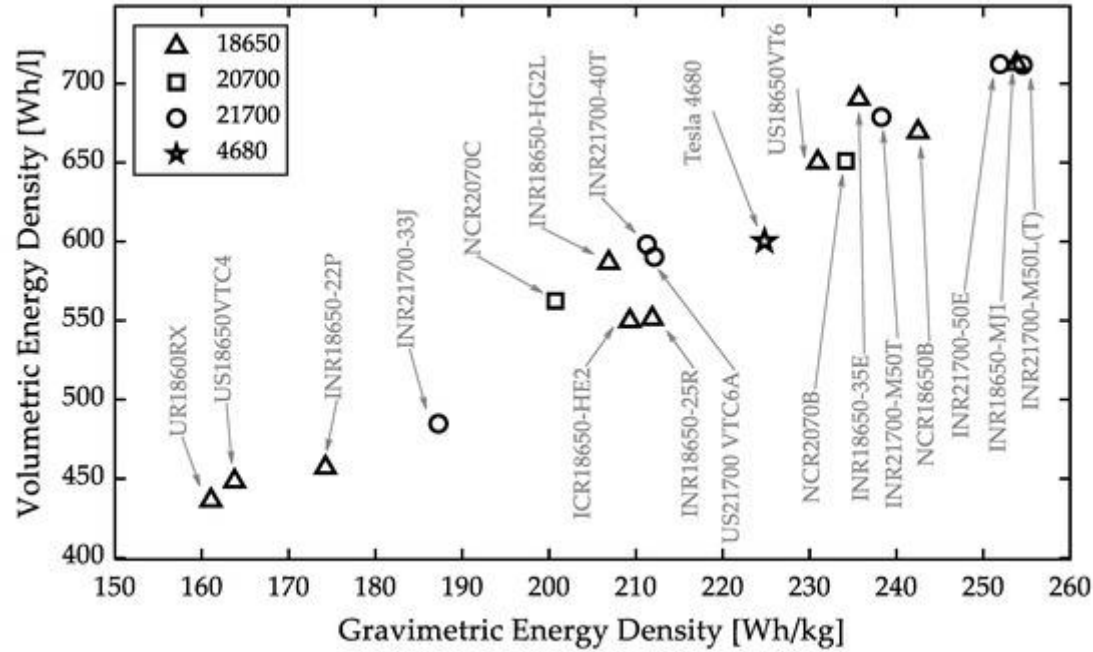
	Myoboc k	i-Limb Quantu m	Hero Arm	Taska Arm	Fillauer	Vincent Systems	Psyonic
Battery Type	Li-ion	Li-poly	Li-ion	---	Li-ion	1C-1P Li-poly	
Capacity	3450mAh	2000mAh	2600mAh	2000mAh	1150mAh	2580mAh *	2200mAh
Voltage	7.4V	7.4V	7.5V	7.4V	7.2V	3.7V	7.4V
Max discharge current	---	5A	---	5.5A	---	6.45A	---
Weight	120g	---	---	88.6g	---	---	---

Link	[1]	[2]	[3]	[4]	[5]	[6]	[7]
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* 2x 1290mAh

Table X – Battery system specifications for current prosthetics

Form factor selection also plays an important role. Cylindrical cells are widely used in applications such as electric vehicles due to their safety, cost-effectiveness, and ease of replacement. However, testing inconsistencies across manufacturers complicates direct comparison of datasheet values. Figures X (doi.org/10.3390/batteries9060309), X, and X (<https://endless-sphere.com/sphere/threads/cycle-life-tests-of-high-energy-density-cylindrical-cells.114486/>, <https://endless-sphere.com/sphere/threads/cycle-life-tests-of-high-power-density-cylindrical-cells.114473/>) show gravimetric and volumetric energy densities of several commercially available cylindrical cells tested under consistent conditions. Few cylindrical cells exceeded 280 Wh/kg, making them less attractive for this application.



<https://doi.org/10.3390/batteries9060309>

18650 size	LG INR18650 M36		Panasonic NCR18650V1		EVE INR18650-33V		Lishen LR18650HB		BAK N18650 CP-35		EVE INR18650 35V	
	New	1000 cycle	New	1000 cycle	New	1000 cycle	New	1000 cycle	New	1000 cycle	New	1000 cycle
0.2C nom. capacity	3361 mAh	2824 mAh	3472 mAh	2639 mAh	3149 mAh	2584 mAh	3349 mAh	3030 mAh	3253 mAh	2899 mAh	3484 mAh	
0.2C nom. energy	12.26 Wh	10.51 Wh	12.61 Wh	9.60 Wh	11.53 Wh	9.65 Wh	12.12 Wh	10.96 Wh	11.71 Wh	10.41 Wh	12.76 Wh	
ACIR 1kHz (40% SoC)		26.80 mΩ		24.42 mΩ		24.15 mΩ		27.32 mΩ		27.26 mΩ	25.00 mΩ	
DCIR 10s (50% SoC)	37.38 mΩ	39.65 mΩ	38.01 mΩ	48.64 mΩ	36.71 mΩ	37.71 mΩ	46.26 mΩ	46.22 mΩ	38.08 mΩ	51.30 mΩ	37.87 mΩ	
DCIR IEC (50% SoC)	31.11 mΩ	33.38 mΩ	29.47 mΩ	38.61 mΩ	30.02 mΩ	31.29 mΩ	35.36 mΩ	37.45 mΩ	30.93 mΩ	41.98 mΩ	30.56 mΩ	
Volumetric ED	717 Wh/l		726 Wh/l		667 Wh/l		683 Wh/l		671 Wh/l		746 Wh/l	
Gravimetric ED	254 Wh/kg		269 Wh/kg		250 Wh/kg		260 Wh/kg		246 Wh/kg		272 Wh/kg	
1C test capacity	2964 mAh	2707 mAh	2934 mAh	2020 mAh	2796 mAh	2425 mAh	2900 mAh	2519 mAh	2745 mAh	2366 mAh	3080 mAh	
1C test energy	10.53 Wh	9.65 Wh	10.44 Wh	7.11 Wh	10.02 Wh	8.75 Wh	10.26 Wh	8.88 Wh	9.78 Wh	8.31 Wh	11.04 Wh	
1C test temp. FC	30.27 °C	33.12 °C	30.03 °C	31.75 °C	28.87 °C	30.51 °C	30.15 °C	32.20 °C	29.26 °C	30.15 °C	29.01 °C	
1C test temp. AC	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	43.32 °C	
2C test capacity	2728 mAh	2549 mAh	2659 mAh	1631 mAh	2651 mAh	2342 mAh	2660 mAh	2158 mAh	2543 mAh	2055 mAh	2905 mAh	
2C test energy	9.37 Wh	8.76 Wh	9.22 Wh	5.55 Wh	9.21 Wh	8.16 Wh	9.11 Wh	7.36 Wh	8.84 Wh	7.01 Wh	10.09 Wh	
2C test temp. AC	38.88 °C	42.44 °C	38.86 °C	41.99 °C	35.18 °C	40.15 °C	38.99 °C	43.14 °C	36.65 °C	39.06 °C	37.26 °C	
3C test capacity	2390 mAh	2251 mAh	2329 mAh	1321 mAh	2456 mAh	2215 mAh	2378 mAh	1810 mAh	2353 mAh	1768 mAh	2679 mAh	
3C test energy	8.02 Wh	7.54 Wh	7.92 Wh	4.36 Wh	8.35 Wh	7.52 Wh	7.93 Wh	5.99 Wh	8.00 Wh	5.88 Wh	9.11 Wh	
3C test temp. FC	48.89 °C	52.04 °C	48.87 °C	51.80 °C	42.71 °C	50.68 °C	49.81 °C	55.70 °C	45.78 °C	48.83 °C	47.56 °C	
3C test temp. AC	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	75.08 °C	

18650 size	Vapcell INR18650 N40		Vapcell INR18650 F38		Vapcell INR18650 F36		Vapcell INR18650 N36		reserved		reserved	
	New	1000 cycle	New	1000 cycle	New	1000 cycle	New	1000 cycle				
0.2C nom. capacity	4022 mAh		3772 mAh		3475 mAh		3534 mAh					
0.2C nom. energy	14.26 Wh		13.54 Wh		12.72 Wh		12.70 Wh					
ACIR 1kHz (40% SoC)	14.51 mΩ		13.84 mΩ									
DCIR 10s (50% SoC)	34.66 mΩ		30.95 mΩ		39.14 mΩ		43.36 mΩ					
DCIR IEC (50% SoC)	23.66 mΩ		22.11 mΩ		31.62 mΩ		34.41 mΩ					
Volumetric ED	833 Wh/l		792 Wh/l		726 Wh/l		725 Wh/l					
Gravimetric ED	298 Wh/kg		288 Wh/kg		271 Wh/kg		272 Wh/kg					
1C test capacity	3294 mAh		3189 mAh		3061 mAh		2960 mAh					
1C test energy	11.56 Wh		11.35 Wh		10.92 Wh		10.37 Wh					
1C test temp. FC	30.83 °C		29.34 °C		29.78 °C		28.69 °C					
1C test temp. AC	41.73 °C		37.66 °C		39.84 °C		41.99 °C					
2C test capacity	3039 mAh		2988 mAh		2848 mAh		2635 mAh					
2C test energy	10.44 Wh		10.40 Wh		9.83 Wh		8.96 Wh					
2C test temp. AC	38.33 °C		37.13 °C		38.86 °C		35.54 °C					
3C test capacity	2822 mAh		2831 mAh		2552 mAh		2275 mAh					
3C test energy	9.47 Wh		9.64 Wh		8.62 Wh		7.57 Wh					
3C test temp. FC	49.76 °C		46.74 °C		49.19 °C		43.09 °C					
3C test temp. AC	>80 °C		73.95 °C		76.94 °C		>80 °C					

test: values measured in the range 4.15-3.00 V which corresponds to ca 90% DoD

FC: values measured in temperature chamber with **Forced Cooling** by airflow with 25 ±0.1 °C

AC: values measured in temperature chamber with **Ambient Cooling** by static air (100l volume) with 25 ±0.1 °C at the beginning of the test (allowed air temperature rise during the test is up to 28 °C)

<https://endless-sphere.com/sphere/threads/cycle-life-tests-of-high-energy-density-cylindrical-cells.114486/>

21700 size	Vapcell INR21700 F60		Lishen LR2170SK		LG INR21700 M58		LG INR21700 M58T		Samsung INR21700 53G(1)		Samsung INR21700 58E	
	New	1000 cycle	New	1000 cycle	New	1000 cycle	New	1000 cycle	New	1000 cycle	New	1000 cycle
0.2C nom. capacity	5965 mAh		5699 mAh		5666 mAh				5186 mAh		5559 mAh	
0.2C nom. energy	21.20 Wh		20.26 Wh		20.37 Wh				18.83 Wh		20.23 Wh	
ACIR 1kHz (40% SoC)	13.81 mΩ		13.83 mΩ		13.45 mΩ				14.40 mΩ		10.77 mΩ	
DCIR 10s (50% SoC)	25.00 mΩ		24.95 mΩ		22.79 mΩ				23.57 mΩ		21.03 mΩ	
DCIR IEC (50% SoC)	18.41 mΩ		19.72 mΩ		18.01 mΩ				17.88 mΩ		15.82 mΩ	
Volumetric ED	809 Wh/l		814 Wh/l		784 Wh/l				747 Wh/l		788 Wh/l	
Gravimetric ED	293 Wh/kg		293 Wh/kg		284 Wh/kg				281 Wh/kg		276 Wh/kg	
1C test capacity	5785 mAh		5496 mAh		5494 mAh				5097 mAh		5416 mAh	
1C test energy	19.88 Wh		18.88 Wh		19.10 Wh				17.81 Wh		19.05 Wh	
1C test temp. FC	33.00 °C		32.18 °C		33.47 °C				32.84 °C		31.82 °C	
1C test temp. AC	51.43 °C		45.61 °C		47.92 °C				45.86 °C		43.25 °C	
2C test capacity	5796 mAh		5440 mAh		5473 mAh				5031 mAh		5416 mAh	
2C test energy	19.25 Wh		18.02 Wh		18.35 Wh				16.92 Wh		18.35 Wh	
2C test temp. FC	42.39 °C		40.97 °C		44.93 °C				44.53 °C		41.60 °C	
2C test temp. AC	>60 °C		>60 °C		>60 °C				>60 °C		>60 °C	
3C test capacity	5660 mAh		5370 mAh		5378 mAh				4954 mAh		5378 mAh	
3C test energy	18.27 Wh		17.31 Wh		17.46 Wh				16.12 Wh		17.72 Wh	
3C test temp. FC	54.60 °C		52.66 °C		59.63 °C				57.77 °C		52.66 °C	
3C test temp. AC	>60 °C		>60 °C		>60 °C				>60 °C		>60 °C	

test: values measured in the range 4.20-2.50 V which corresponds to ca 100% DoD

FC: values measured in temperature chamber with **Forced Cooling** by airflow with 25 ±0.1 °C

AC: values measured in temperature chamber with **Ambient Cooling** by static air (100l volume) with 25 ±0.1 °C at the beginning of the test (allowed air temperature rise during the test is up to 28 °C)

DNF: Did not finish, cell nominal capacity drop under 70 % SoH before 1000 cycles, N/A - not applicable, TBD - To be determined, ED - Energy Density, SoC - State of Charge, IEC - according to IEC 61960-2330

<https://endless-sphere.com/sphere/threads/cycle-life-tests-of-high-energy-density-cylindrical-cells.114486/>

In contrast, pouch cells generally achieve higher gravimetric energy densities for the same chemistries and are more common for newer battery technologies because of their manufacturability. The best available pouch cells that were found are from Amprius and offer energy densities ranging from 360 to 380 Wh/kg. A discussion with a sales representative revealed that there will be a next-generation line opening in Q4. This series is expected to reach 430 Wh/kg based off a pilot production line. Based on this analysis, pouch cells are the preferred choice for the prosthetic arm.

Subsystem 1.3.2 – Battery Management System (BMS)

A Battery Management System (BMS) is essential for prolonging battery life and ensuring safe operation. Numerous circuit topologies exist for balancing cells (doi.org/10.3390/batteries9030152), ranging from simple passive balancing to more sophisticated active methods. However, the prosthetic arm’s modest charge and discharge rates impose relatively low stress on the battery. In addition, the operational environment will generally remain within human-safe temperature and humidity ranges.

Given these factors, a simple diode-based BMS with temperature monitoring is sufficient for the application. A proof-of-concept BMS was developed for a 14S6P battery pack, producing a nominal output of 51.8 V—an appropriate starting voltage for subsequent boost conversion to the high voltages required by the DEAs.

A separate PCBA was created that can add features for a hot-swappable battery. See Figure X.

Table 1. Comparison of the charge balancing topologies [4,53,63].

Balancing Techniques, Methods, and Types	No. of Elements for Balancing (n Cells)	Balancing Time, Control Complicity	Power Loss, Efficiency	Voltage and Current Stress	Size and Cost	Benefits	Drawbacks
Fixed Shunt, Passive and Fixed	n resistors	Slow, Very Simple	Very High, Poor	Zero/Zero	Very Small, Very Cheap	Very simple control system, very small size and cheap	Long balance time, high power loss, require thermal management, poor efficiency
Switch Shunt, Passive and Only Charging	n switches, n resistors	Slow, Simple	Very High, Low	High/High	Very Small, Very Cheap	Simple control system, vary cheap and small in size, suitable to apply in HEV but face some limitations for applying in EV	Long balance time, high power loss, require thermal management, poor efficiency
Analog Shunt, Passive, and Only Charging	n switches, n Op-amps, 3n resistors, n capacitors	Slow, Simple	High, Low	High/High	Very Small, Cheap	Simple control system, very small and cheap	High power loss, require thermal management, poor efficiency
Single-Switch Capacitor, Active and Charge/Discharge	n + 5 switches, 1 capacitor	Medium, Complex	Minor, Better	Low/Low	Small, Medium	Bidirectional, simple control, good efficiency, suitable for application in HEV and EV	Control system is complex, and minor power loss
Switch Capacitor, Active and Charge/Discharge	2n switches, n – 1 capacitor	Medium, Medium	Minor, Better	Low/Low	Medium, Medium	Bidirectional, simple control, low current and voltage stress	Many switches needed, medium equalization speed
Double-Tiered Switch capacitor, Active and Charge/Discharge	2n switches, 2n – 3 capacitor	Medium, Complex	Minor, Better	Low/Low	Medium, Medium	Bidirectional, good efficiency, fast balancing compared with switch capacitor	Many switches needed, medium equalization speed
Modularized Switch Capacitor, Active and	M(n + 2) switches, ...	Medium, ...	Minor, Better	Low/Low	Medium, ...	Bidirectional, low current and voltage stress, applied in high power	Requires many switches, complex control

Table X – Incomplete table from <https://doi.org/10.3390/batteries9030152>

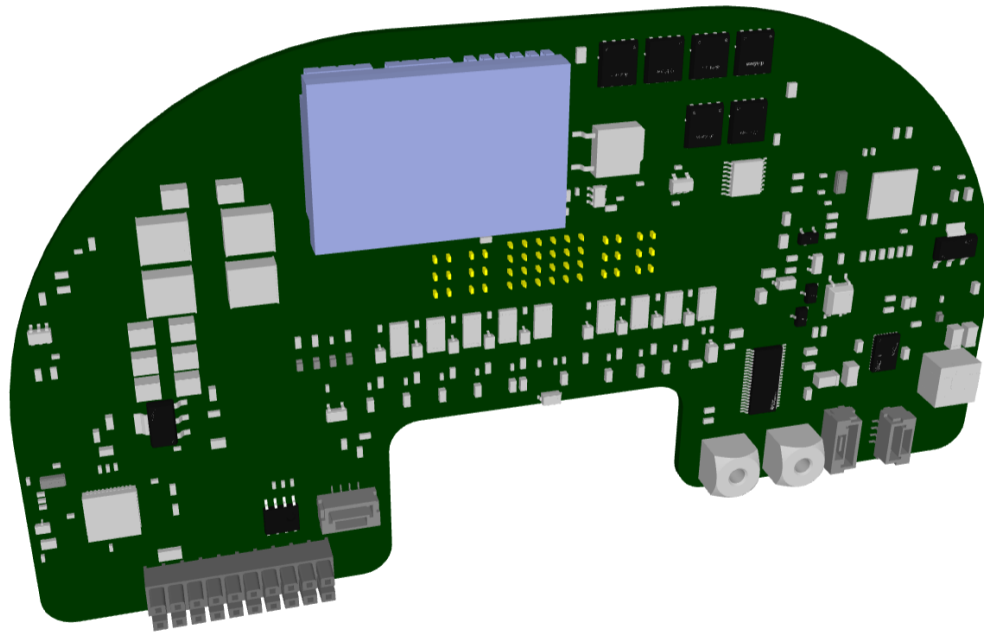


Figure X – Battery Management System PCBA

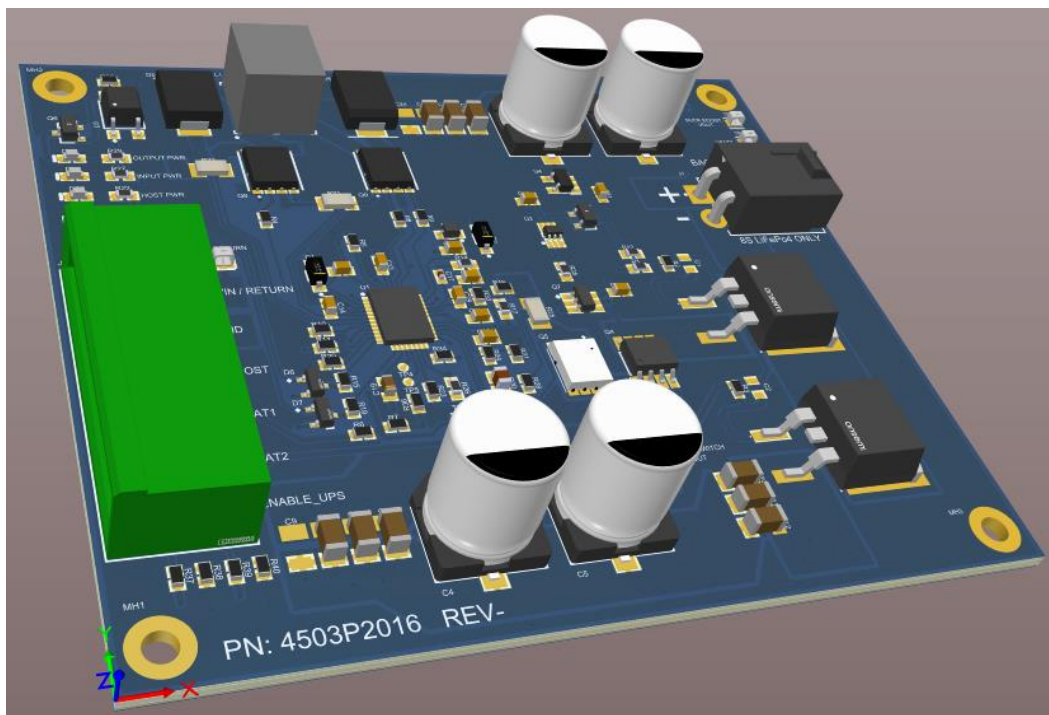


Figure X – Hot swappable battery PCBA

Subsystem 1.3.3 – Power Train

One major challenge in DEA systems is their requirement for high operating voltages. Fortunately, despite the high voltage, the overall power consumption remains moderate, allowing a conventional power train topology with careful component selection for high-voltage reliability.

Two proof-of-concept printed circuit board assemblies (PCBAs) were developed: one serving as a command board and the other as a power conversion board. Separating these functions enabled easier debugging and modular upgrades, though they could be combined in future versions. The power conversion board uses an isolated buck-boost DC-DC converter with a high duty cycle to modulate the BMS output voltage, boosting it efficiently with minimal energy loss. The command board uses I²C to communicate with multiple slave ICs that monitor output voltage, output current, converter status, and system control functions.

The boards were designed following guidelines from IEC 61010-1, a safety standard for medical and laboratory equipment. Additional safety precautions were implemented, including the use of redundant, functionally safe microcontrollers to monitor critical parameters. In the event of a failure in the primary control pathway, the system is designed to safely shut down and request maintenance intervention.

While the current PCBAs meet basic functionality, they lack dedicated math acceleration ICs that could offload computational tasks. Future iterations could integrate these features once the software architecture is more fully developed.

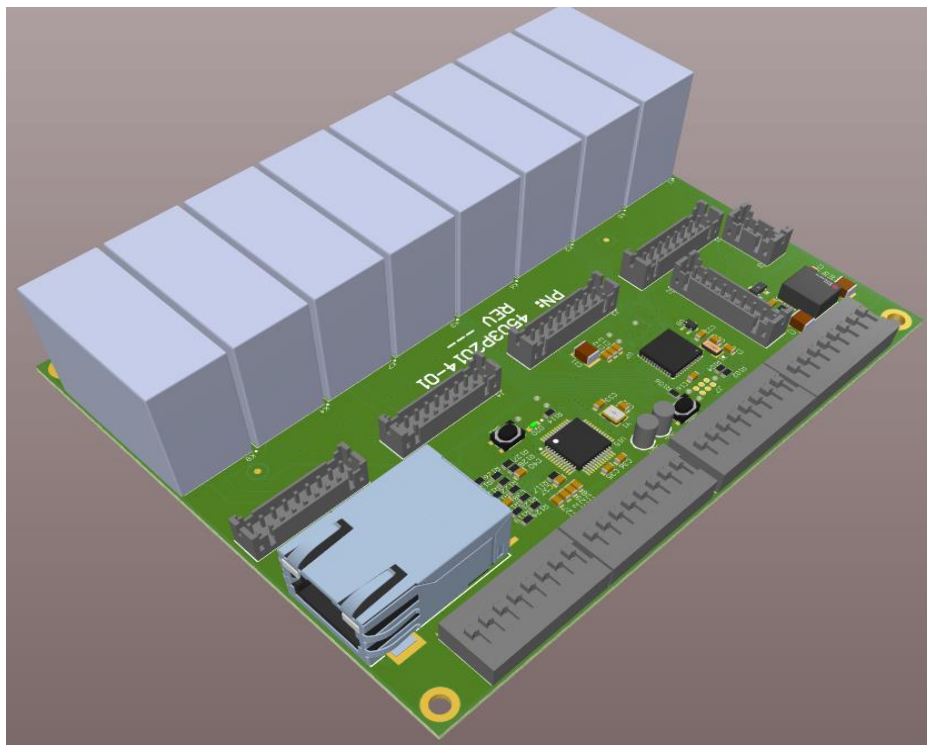


Figure X – Command PCBA

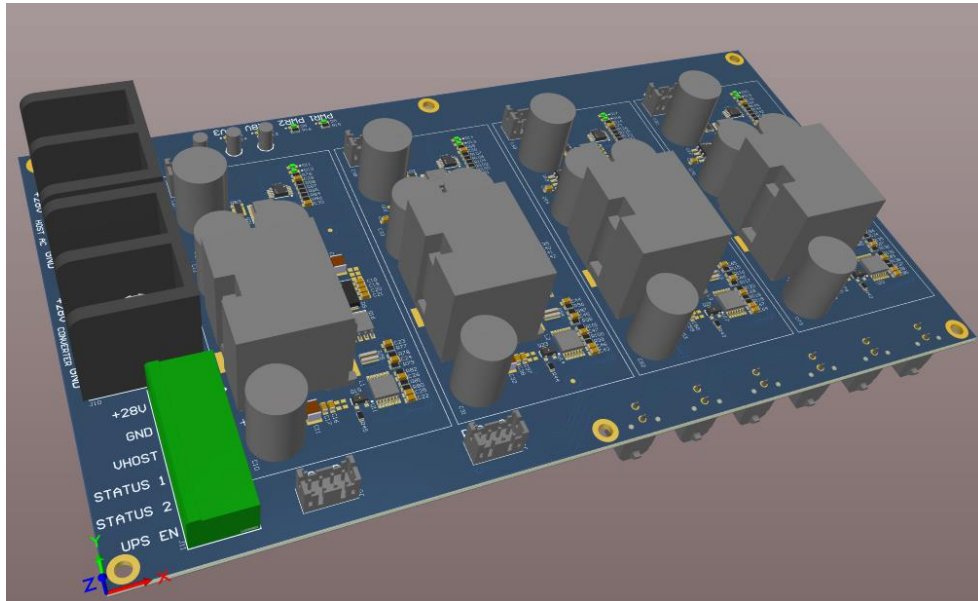


Figure X – Four independently controlled boost converters on one PCBA

Subsystem 1.3.4 – DEA Control and Energy Recovery

This subsystem is still under active development. Preliminary design considerations include splitting system controls into two categories: critical (e.g., large actuators for elbow, wrist, or universal grasp) and non-critical (e.g., user interface, low-power monitoring). Critical paths will require higher reliability and stricter safety protocols.

Final design decisions for this subsystem are dependent on the selection and specification of the DEA hardware. Discussions with Elysium Robotics in February 2025 indicated interest in supporting prototype DEA development if further progress is demonstrated. A working power train and sensing/control proof of concept may satisfy these requirements.

Subsystem 1.3.5 – Microcontrollers

The system uses four microcontrollers to manage operations. A SAM32 microcontroller (ARM Cortex-M4 core) handles general decision-making, non-critical functions, and coordination with a second microcontroller (WIZ5500) responsible for parsing Ethernet messages. Two additional ATSAM21D microcontrollers are dedicated to monitoring and controlling safety-critical functions through redundant pathways. This redundancy ensures that even in the event of a failure in the primary system, the prosthetic remains safe and can appropriately alert the user for maintenance.

3.4 – Sensors

Accurate and reliable control of a prosthetic limb requires more than a single type of signal. While surface electromyography (sEMG) provides the primary control input, it is sensitive to numerous external and internal factors such as skin impedance changes, muscle fatigue, environmental conditions, and user movement patterns. To build a robust, adaptable system, additional sensing modalities are incorporated to monitor physiological states, environmental conditions, and mechanical interactions. This multi-sensor approach not only improves system fault tolerance but also enables more advanced control strategies, such as dynamic grip adjustment and long-term adaptation to user-specific anatomical changes. This section outlines the sensors included in the prosthetic arm and explains the role each sensor plays in supporting high-fidelity control.

The primary sensor array is dedicated to surface electromyography (sEMG). Following a structure similar to the TASKA arm, the array is designed as a high-density circular grid placed consistently around the upper arm, rather than being selectively positioned over specific muscles of interest. This high-density configuration creates a continuous spatial map of muscle activity, while also offering redundancy; individual sensors with faults, poor signal quality, or damage can be excluded from processing without significantly degrading overall performance.

To complement the sEMG system, a galvanic skin response (GSR) sensor is included. This sensor monitors changes in skin impedance caused by factors such as sweating, which can affect sEMG signal quality. By tracking GSR, the system can apply corrections or compensations to help maintain reliable signal interpretation under varying environmental and physiological conditions.

Additionally, a sonomyography-based sensor array (doi.org/10.1109/TUFFC.2023.3333318, <https://doi.org/10.1109/TNSRE.2019.2953588>,) is included to monitor anatomical changes over time, such as swelling during exercise, hydration variations, or muscle hypertrophy. Small, low-frequency ultrasound sensors are a likely candidate for this application. It is important to note that these sensors are intended for long-term trend monitoring, not for real-time primary motion control, in contrast to some experimental systems that use sonomyography as the main input.

A heartbeat monitor is also incorporated. These sensors are small, inexpensive, and easily integrated. The primary purpose is to help identify and potentially remove heartbeat-related noise patterns from the sEMG signal. If this signal proves to have minimal utility, it may be omitted in future design iterations.

An inertial measurement unit (IMU) (<https://invensense.tdk.com/products/motion-tracking/9-axis/>) is mounted to track the limb's position and motion. The IMU will provide

acceleration data for use in reverse kinematics calculations that estimate joint angles and limb dynamics. Additionally, the IMU can help detect external forces applied to the forearm, such as the load from carrying a bag, which might not be directly captured by grip force sensors.

Finally, force sensors are integrated into the hand and fingers to assist with gripping tasks. In the absence of full sensory feedback to the user, these sensors allow the control software to adjust grip strength dynamically—tightening or relaxing the grip as necessary to prevent object slippage or damage.

For validation and verification, research-grade sensors like <https://www.noraxon.com/our-products/ultium-emg/>, <https://delsys.com/sensors/>, <https://polhemus.com/motion-tracking/hand-and-finger-trackers/> will be helpful by providing a higher resolution and sampling rate.

3.5 – Software and Control Algorithms

The software architecture for the prosthetic arm must support a wide range of functions, spanning low-level hardware management and high-level control algorithms. At the system management level, the software must monitor battery health, optimize charging behavior, and report key diagnostic information to the user. At the motion control level, it must interpret signals from the sEMG sensor array, filter and denoise the data, and implement a control pipeline capable of translating user intent into natural, coordinated limb movements. Special attention is given to the development of an adaptive observer, capable of inferring internal motor states from noisy surface signals, and a non-linear controller, capable of handling the irregular and complex nature of upper limb motion. The following sections describe the software components in more detail, from battery management to high-level intent decoding and control execution.

Battery Management and Health Monitoring

The software must include functionality dedicated to battery charging control and battery health monitoring. For charging, a conventional constant-current, constant-voltage (CC-CV) algorithm will likely be sufficient. Assuming that the user charges the device overnight, the constant-current (CC) phase charging current should be dynamically adjusted based on the time available until the expected wake-up time, minus a predefined buffer period allocated for the constant-voltage (CV) phase. For example, if the user typically wakes at 6:00 a.m. and charging begins at 2:00 a.m., the system would select a faster charging rate than if charging began at 10:00 p.m.—while always respecting the battery manufacturer's recommended charging parameters.

If it is known, for example, that the CV phase requires approximately 45 minutes to bring the battery from 90% to 100% state of charge, the system would aim to complete the CC phase by 5:15 a.m. This approach maximizes battery lifespan by using the longest possible charging time within operational constraints.

Battery state-of-health (SOH) monitoring will be implemented using coulomb counting during charge and discharge cycles. Accurate SOH estimation will allow the system to provide maintenance recommendations to the user, such as suggesting battery replacement before unexpected performance degradation occurs. It also helps accurately estimate the capacity of the battery, which assists the charging algorithm.

Actuation Control

Motion control will rely on standard reverse kinematics equations to translate target limb positions into joint movements, based on input from the sEMG observer. Special consideration will be given to compensating for non-linear behavior in the actuators, particularly the DEAs. To enhance the natural appearance of movements and reduce mechanical harshness, a motion controller will be implemented, optimizing angular accelerations at the joints during movement transitions to prevent jerky, sudden starts and stops.

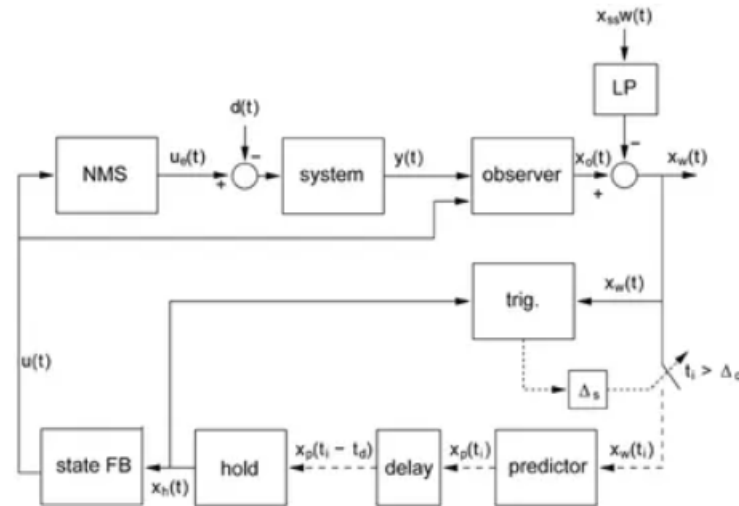
sEMG Observer and Non-Linear Control Framework

More sophisticated sections of the software revolve around the sEMG signal observer and the non-linear controller responsible for decoding intent and executing complex motion patterns. Unlike the relatively regular dynamics of gait or quiet standing, upper limb motion is highly irregular and cannot be easily modeled as a limit cycle.

In locomotion control research, for example, human walking has been successfully modeled by defining an idealized gait trajectory parameterized by a single phase variable. Controllers stabilize deviations from the ideal gait using techniques such as limit cycles and Lyapunov-based stabilization (doi.org/10.1109/TAC.2017.2648040, doi.org/10.1109/TAC/2014.2299335, doi.org/10.1109/TNNLS.2016.2584559). Quiet standing control, particularly in hip-based balance strategies, similarly defines an equilibrium point (center of mass over the base of support) and uses energy-minimization principles to maintain posture (doi.org/10.1016/j.humov.2007.11.005).

However, upper limb tasks—such as daily grooming, dressing, or eating—require frequent, irregular, and context-dependent movements. There is no simple cyclic reference motion to stabilize against. Therefore, upper limb intent decoding must be broad, adaptive, and computationally richer than controllers used for walking or standing balance. This challenge motivates the inclusion of a multi-modal sensor array to improve the quality and reliability of sEMG signal interpretation.

After denoising the sEMG signals and extracting relevant features, the design of the control pipeline can vary substantially depending on system goals and available computation. As a starting point, Figure X shows the recommended processing pipeline adapted from doi.org/10.3389/fncom.2013.00055. Regardless of the specific signal processing method selected, all pipelines converge on the need for an adaptive observer.



doi.org/10.3389/fncom.2013.00055

The adaptive observer compensates for the fact that the prosthesis cannot directly measure internal motor intent. sEMG signals reflect heavily processed outputs of the central nervous system, already filtered through layers of motor unit recruitment strategies. As such, an observer is required to infer the underlying "plant" state based on these partial, noisy signals.

A non-linear observer capable of accounting for these characteristics can likely be synthesized from techniques described in the research papers below. Observer design will focus on stability, adaptability to signal drift, and robustness against external disturbances.

doi.org/10.1049/iet-cta.2015.0092

doi-org.libdata.lib.ua.edu/10.1002/rnc.3543

doi.org/10.1016/j.automatica.2009.04.002

doi.org/10.1016/j.automatica.2014.10.068

doi.org/10.1109/TAC.2018.2839526

doi.org/10.1016/j.cnsns.2023.107234

Controller Synthesis

Following intent estimation by the observer, the prosthetic controller must convert the decoded internal commands into limb motions. Control synthesis will leverage the W and C matrices introduced in Section 2.3, representing spatial and temporal synergies, respectively. By incorporating the non-linearities captured across the W and C matrices along with real-time state estimates, the system can generate appropriate acceleration profiles for each joint. These profiles can then be integrated to compute position and velocity commands for the actuators, enabling naturalistic, efficient movement across a wide range of tasks. A good reference is doi.org/10.1109/TIE.2016.2538741.

~~3.5 – Dielectric Actuator Coordination via Synergy Models~~

~~3.6 – Implementation and Integration~~

~~4 – Evaluation and Comparison~~

~~4.1 – Performance Metrics Against Human Benchmarks *~~

~~4.2 – Comparison with Existing Prosthetics *~~

~~4.3 – Limitations and Observations *~~

~~5 – Future Considerations and Development Pathways~~

~~5.1 – Improving Control Robustness *~~

~~5.2 – Integration with Sensory Feedback~~

Receptor	Fiber type	Sensation	Adaptation	Receptive field	Perceptual functions
Pacinian corpuscle	A _β	Vibration	RA2		Vibratory cues transmitted by body contact
Meissner's corpuscle	A _β	Touch, flutter, stretch	RA1		Skin motion, detecting slipping motion
Merkel's disc / receptor	A _β	Touch, pressure	SA1		Fine tactile discrimination, form and texture perception
Ruffini's corpuscle	A _β	Stretch	SA2		Skin stretch, direction of object motion, hand shake
Free nerve endings	C A _δ	Temperature	SA		Cold and warm temperature, Pain
Hair receptor	A _β	Hair movement	RA1		Skin movement

~~5.3 – Scalability and Real-world Deployment *~~

6 – Conclusion

7 – References

8 – Appendixes

Guiding Body	Program Name	Start Year	End Year	Website
VA	PSAS - Prosthetic and Sensory Aids Service	2000	Ongoing	https://www.prosthetics.va.gov/psas/index.asp
DARPA	RP - Revolutionizing Prosthetics	2006	2016 (transitioned)	https://www.darpa.mil/research/programs/revolutionizing-prosthetics
NIH	BRAIN Initiative	2013	Ongoing	https://braininitiative.nih.gov/
NSF/NIH	SCH - Smart and Connected Health	2013	Ongoing	https://www.nsf.gov/funding/opportunities/sch-smart-health-biomedical-research-era-artificial-intelligence/nsf18-541/solicitation
DARPA	HAPTIX	2014	2019 (transitioned)	https://www.darpa.mil/research/programs/hand-proprioception-and-touch-interfaces
NIH	SPARC	2015	Ongoing	https://commonfund.nih.gov/sparc
DoD / CDMRP	OPORP - Orthotics and Prosthetics Outcomes Research	2015	Ongoing	https://cdmrp.health.mil/opor/default
DARPA	NESD - Neural Engineering System Design	2016	Ongoing	https://www.darpa.mil/research/programs/neural-engineering-system-design
NSF	EFRI - Engineering the Brain	2016	Ongoing	https://www.nsf.gov/eng/efma/emerging-frontiers-research-innovation
USAF / DoD	AFWERX Innovation Challenges	2017	Ongoing	https://afwerx.com/
DARPA	N3 - Next-Generation Nonsurgical Neurotechnology	2018	Ongoing	https://www.darpa.mil/research/programs/next-generation-nonsurgical-neurotechnology
NIH	RE-JOIN	2022	Ongoing	https://heal.nih.gov/research/preclinical-translational/restoring-joint-health-function-reduce-pain
ARPA-H	ARPA-H	2022	Ongoing	https://arpa-h.gov/

Commercial Products				
Manufacturer	Product	DoF	Grips	Link
Unlimited Tomorrow	TrueLimb	---	6	https://www.unlimitedtomorrow.com/truelimb/
Open Bionics / Open Bionics Lab	Hero Arm	4	6	https://openbionics.com/
Ossur	i-limb	7	36	https://www.ossur.com/en-us
Deka Research (Mobius Bionics)	Luke Arm	10	2	https://mobiusbionics.com/
Fillauer LLC	Utah arm, TASKA	---	20	https://fillauer.com/
HDT Global	Military Prosthetics	---	---	https://www.hdtglobal.com/product/prosthetics/
BrainCo	Neuromaker	---	---	https://www.neuromakerstem.com/
Cowi	Cowi Arm	6	24	https://www.cowi.com/
Alt-Bionics	Genesis	6	0	https://www.altbionics.com/prosthetics/
Proteor (Freedom Innovations)	Quatro, Allux, Plie	Lower limb		https://us.proteor.com/
Blatchford Inc. (Endolite)	Linx, Echelon, Elan	Lower limb		https://www.blatchfordmobility.com/en-us/
Lifeward (ReWalk Robotics)	Rewalk	Exoskeleton		https://golifeward.com/
Palladyne AI (Sacros)	Guardian XO	Exoskeleton		https://www.palladyneai.com/
Esko Bionics	Esko EVO	Exoskeleton		https://eksobionics.com/
Myomo	MyoPro	Exoskeleton		https://myomo.com/

PN	ICP521630PM-01	NCA463436A	LP502030	LP472040F
Manufacturer	Renata	Panasonic	EEMB	EEMB
Type	LiPo	Lithium-ion*	Lithium-ion	LiFePO4
Capacity	250mAh	720mAh	250mAh	200mAh
Nominal Voltage	3.7V	3.6V	3.7V	3.2V
Charge Current	125mA	---	250mAh	200mAh
Discharge Current	250mA	---	500mAh	400mAh
Weight	5.4g	12.4g	5.3g	5.0g
Volume	5.7 x 17 x 31.5mm	4.6 x 34.3 x 35.5mm	5.3 x 20.5 x 31mm	5 x 20.5 x 41mm
Internal Impedance	290mOhm	---	190mOhm	250mOhm
Cycles	---	---	500-800	2000
Cost	\$16.30	---**	\$7.92	---**
Link	[8]	[9]	[10]	[11]

* Advertised as NNP and “New Nickel Oxide Based Platform” but seems to be LiM_xO_{n2}

** Not sold individually

Table 2 – Commercial pouch batteries in 250mAh range